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# Circuit Fingerprints: How Answer Tokens Encode Their Geometrical Path

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## Abstract

Circuit discovery and activation steering in transformers have developed as separate research threads, yet both operate on the same representational space. Are they two views of the same underlying structure? We show they follow a single geometric principle: answer tokens, processed in isolation, encode the directions that would produce them. This *Circuit Fingerprint* hypothesis enables circuit discovery without gradients or causal intervention—recovering comparable structure to gradient-based methods through geometric alignment alone. We validate this on standard benchmarks (IOI, SVA, MCQA) across four model families, achieving circuit discovery performance comparable to gradient-based methods. The same directions that identify circuit components also enable controlled steering—achieving 69.8% emotion classification accuracy versus 53.1% for instruction prompting while preserving factual accuracy. Beyond method development, this read-write duality reveals that transformer circuits are fundamentally geometric structures: interpretability and controllability are two facets of the same object.

## 1. Introduction

Mechanistic interpretability seeks to understand neural networks by identifying the components responsible for specific computations. Two approaches have emerged: *circuit discovery*, which locates important components through causal intervention or gradient approximation (Meng et al., 2022; Wang et al., 2023; Nanda, 2023; Kramár et al., 2024), and *activation steering*, which controls model behavior by intervening along learned directions (Turner et al., 2023; Zou et al., 2023). Despite operating on the same components in the same representational space, these approaches have been studied separately.

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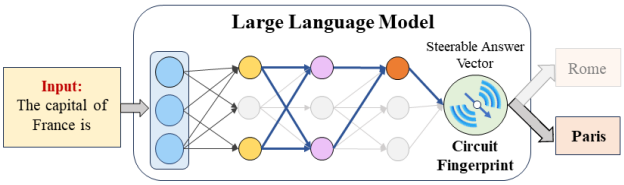


*Figure 1. Circuit Fingerprints* unifies circuit discovery and activation steering as dual operations—reading and writing—on the same geometric structure encoded in answer token representations.

We show they are two views of the same underlying structure. Answer tokens—the outputs a model produces—encode not just semantic content but the computational pathways that generated them (Fig. 1). The representation of "Paris" carries a geometric signature of the circuit that would produce it, regardless of context. We find that this signature can be *read* to identify circuit membership or *written* to steer model behavior—the same directions accomplish both, confirming they reflect genuine computational structure.

This finding connects circuit analysis to the linear representation hypothesis (Park et al., 2024; Elhage et al., 2022): if features are directions in activation space, then circuits manipulating those features should be identifiable through directional alignment. Our results provide direct evidence for this connection, suggesting that transformer circuits are fundamentally geometric structures encoded in the model's activation space.

We validate this hypothesis on three tasks: Indirect Object Identification (IOI) (Wang et al., 2023), Subject-Verb Agreement (SVA) (Marks et al., 2025), and Multiple-Choice Question Answering (MCQA) (Mueller et al., 2025). Our geometric method is comparable ot gradient-based circuit structure without backpropagation, and the same directions enable steering that outperforms instruction prompting in emotion steering (69.8% vs 53.1% emotion accuracy) while preserving factual grounding. This unification simplifies the interpretability toolkit: rather than separate methods for discovery and control, a single geometric analysis reveals both which components matter and how to manipulate them.

**Contributions.**

1. We show that circuit membership can be read directly from geometric alignment with answer token directions, recovering comparable circuit structure to gradient-based methods without requiring backpropagation.

2. We demonstrate that the same directions enable controlled steering—read and write are dual operations on the same geometric structure—validating that we manipulate genuine computational pathways rather than superficial correlations.

3. We show that feature circuits can be discovered by extracting directions from instruction-modified prompts, without task-specific datasets.

## 2. Related Works

**Circuit Discovery.** Identifying sparse subnetworks responsible for specific behaviors is central to mechanistic interpretability (Elhage et al., 2021; Olsson et al., 2022). The activation patching estimates causal importance by replacing activations with counterfactual values (Vig et al., 2020; Meng et al., 2022; Wang et al., 2023). While providing strong causal guarantees, this requires $O(LH)$ forward passes for systematic discovery. Attribution patching (Nanda, 2023) and its integrated gradients variant EAP-IG (Syed et al., 2024) reduce cost via gradient approximation (Mueller et al., 2025). However, gradient-based methods can suffer from saturation and sensitivity to LayerNorm non-linearities (Kramár et al., 2024; Hanna et al., 2024). Optimization-based approaches frame discovery as mask learning (Conmy et al., 2023; Bhaskar et al., 2024), scaling to larger models but requiring iterative optimization.

**Activation Steering.** Representation engineering methods control model behavior by adding learned vectors to activations (Turner et al., 2023; Rimsky et al., 2024; Zou et al., 2023). These methods require contrastive data collection, direction learning, and intervention tuning. Recent work studies steering at specific circuit locations (Todd et al., 2023), but the connection between where to steer and which directions to use remains underexplored.

**Linear Representations.** The linear representation hypothesis posits that semantic features correspond to directions in activation space (Park et al., 2024; Elhage et al., 2022), with empirical support across spatial, temporal, and conceptual features (Gurnee & Tegmark, 2024; Nanda et al., 2023). Building on this, we show that if task features are geometrically encoded, their generating circuits can be found via alignment and controlled via the same directions.

## 3. The Circuit Fingerprint Hypothesis

Circuits in transformers are not constructed dynamically for each input. They are stable structures encoded in the model's weights, activated whenever relevant computation is required (Olsson et al., 2022; Wang et al., 2023). The same components that identify indirect objects operate whether the names are "Mary and John" or "Alice and Bob." The same components that recall capitals operate whether the query concerns France or Germany.

This stability has a surprising implication. Consider two prompts:

*"The answer is Paris. The capital of Italy is ___"*
*"The capital of France is ___"*

In the first prompt, "Paris" appears as an input token unrelated to the query. Yet as it passes through the model, it traverses the same computational pathways responsible for *producing* "Paris"—and may even suppress the model from outputting "Rome" as the answer. The circuit for capital-city recall exists in the weights; "Paris," whether as input or output, engages this structure.

*Hypothesis* 3.1 (Circuit Fingerprint). Answer tokens, processed in isolation, trace the computational pathways that produce them. A component belongs to the circuit distinguishing $a^+$ from $a^-$ if and only if processing these tokens activates the component differentially.

Following work on linear probes (Alain & Bengio, 2017; Belinkov, 2022), steering vectors (Turner et al., 2023; Zou et al., 2023), and sparse autoencoders (Bricken et al., 2023; Cunningham et al., 2024), we assume that features are linearly encoded in activation space (Park et al., 2024; Elhage et al., 2022).

### 3.1. Circuits as Computational Graphs

Transformer computation forms a directed acyclic graph where nodes are model components (attention heads and MLPs) and edges represent information flow through the residual stream. A *circuit* is the subgraph causally responsible for a given task (Elhage et al., 2021; Conmy et al., 2023). Traditionally, circuit membership is established through intervention: a component belongs if ablating it degrades task performance (Meng et al., 2022; Wang et al., 2023).

The Circuit Fingerprint Hypothesis suggests an alternative characterization: circuits are not merely causal structures but *geometric* ones, encoded in the linear structure of activation space. If correct, circuit membership can be determined from inner products with answer token directions rather than from interventions or gradient approximations.

We validate this hypothesis along two axes: *circuit recovery* (do geometric scores approximate causal importance?) and

*causal control* (can the extracted directions steer model outputs?).

## 4. Reading: Geometric Attribution

The Circuit Fingerprint Hypothesis implies a simple approach to circuit discovery: if answer tokens trace the same pathways that produce them, then the directions separating contrastive answers should identify circuit components. We extract these directions, then measure which components align with them.

### 4.1. Extracting Target and Prompt Directions

Our method uses two sources of information: *answer tokens* and *contrastive prompts*. Answer tokens ($a^+$, $a^-$) are processed in isolation to extract target directions—the geometric signature of what distinguishes correct from incorrect outputs. Contrastive prompts (clean and corrupted, e.g., "The capital of France is" vs "The capital of Italy is") are processed to extract activation differences—how the model's internal state changes between conditions. Circuit components are identified by measuring alignment between these two: components whose prompt-induced activation differences align with answer-derived target directions are those contributing to the task.

We use a contrastive approach to isolate the features associated with any pair of answers and their prompts. Given answer tokens $a^+$ and $a^-$—e.g., "Paris" and "Rome"—we pass each through the model in isolation and compute their difference at the last layer:

$$\Delta\mathbf{r}^{(L)} = \mathbf{r}_{a^+}^{(L)} - \mathbf{r}_{a^-}^{(L)} \tag{1}$$

This difference $\Delta\mathbf{r}^{(L)}$ defines the direction in activation space that distinguishes the two answers at the last layer $L$. Any component contributing to the distinction must write along this direction; any component orthogonal to it is irrelevant to the task.

Two forward passes suffice to extract target directions at all layers. For finer-grained analysis, we also extract channel-specific directions in the native spaces of each attention component:

$$\Delta\mathbf{v}^{(\ell,h)} = \mathbf{v}_{a^+}^{(\ell,h)} - \mathbf{v}_{a^-}^{(\ell,h)} \quad \text{(value vectors)} \tag{2}$$

$$\Delta\mathbf{q}^{(\ell,h)}, \Delta\mathbf{k}^{(\ell,h)} = \dots \quad \text{(query/key vectors)} \tag{3}$$

The 3 vector differences are obtained from the prompts as this will be used later to measure the relative importance of edges towards the inputs.

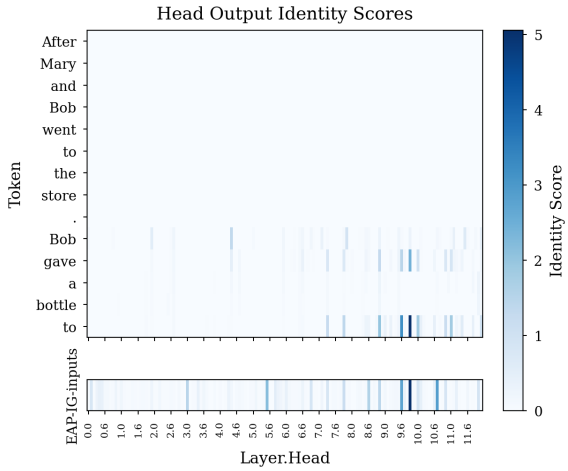


*Figure 2.* Comparison of attention head importance on the IOI task. **Top:** Per-token identity scores computed with respect to answer token's attention head outputs (our method). **Bottom:** Head importance from EAP-IG-inputs (gradient-based). Both methods identify the same critical heads in layers 9–11, with strongest signal at the final token position.

### 4.2. Measuring Component Direct Contributions

The target direction $\Delta\mathbf{r}^{(L)}$ captures the features necessary to distinguish the two answers. We use this direction to measure how much each component contributes to the answer distinction—without requiring interventions or gradient computation.

A naïve approach projects each component's output into the residual stream and measures alignment with $\Delta\mathbf{r}^{(L)}$. However, this introduces noise: the projection matrices ($W_O$ for attention, $W_{out}$ for MLPs) mix the component's internal computation with the shared residual stream geometry.

Instead, we measure alignment in each component's *native space*. For a component $c$ with projection matrix $W_c$ (where $W_c = W_O$ for attention heads and $W_c = W_{out}$ for MLPs), we first transform the target direction into the component's native space:

$$\hat{\mathbf{t}}_c = \frac{W_c^\top \Delta\mathbf{r}^{(L)}}{\|\Delta\mathbf{r}^{(L)}\|} \tag{4}$$

We measure each component's task-relevant contribution as its alignment with this transformed target:

$$S_c = \langle \Delta\mathbf{o}_c,\ \hat{\mathbf{t}}_c \rangle \tag{5}$$

where $\Delta\mathbf{o}_c$ is the differential output of component $c$ when processing the contrastive prompts. Critically, this decomposition preserves additivity: $\sum_c S_c$ equals the total projection onto the target direction in the residual stream space, ensuring component contributions sum to the full effect.

As an initial observation of the contribution of this projections, consider Fig. 2. For the IOI prompt "After Mary and